INVENTORY

A field-by-field explanation of the Inventory page, its tenant CSV import, stock meanings, and how supply changes an MRP plan.

WHAT IS CONNECTED TODAY?

The inventory list and CSV import use tenant database data. Add Inventory, the history tabs, movement chart, and several detail fields are sample UI and do not persist or load transaction history.

THE FLOW

MRP explodes forecast through the bill of materials, adds expected receipts, consumes projected stock, and recommends replenishment when stock would fall below its minimum.

Page map and summary cards

Tenant-backed list and importer, alongside controls that are still demonstrative.

HEADER AND SUMMARY

Tenant selector	Selects the tenant database context. Confirm this before upload.
Search	Client-side search across material/plant code, name, and category.
Filters	Notification only; no filter panel or server query is connected.
Upload type + Upload CSV	Inventory & stock CSV import; accepts .csv up to 10 MB. Success reloads list.
Add Inventory	Opens a sample form. Submit updates browser state only, not database.
Total Items	Count of API rows. Active material/plant pairs can show zero even without inventory record.
Total Inventory Value	Sum of current stock x material unit_cost; illustrative INR formatting.
Low Stock Items	Count where server status evaluates available quantity below minimum.
Out of Stock Items	Count where on-hand quantity is zero.
vs last month / arrow	Decorative fixed comparison; no historical comparison is calculated.

HOW THE API BUILDS THE LIST

GET /api/inventory returns active material x active plant combinations in the selected tenant. Missing inventory records are left-joined and shown as zero stock. An optional plantCode filter exists in the API but is not exposed on this screen. A failed request is caught as an empty list, so confirm logs/network before assuming the tenant has no stock.

Tabs and table columns

The current status calculation and the limits of each tab.

TABS

All Items	All returned rows; works.
Low Stock	Rows with API Low Stock status; works.
Out of Stock	Rows with zero on-hand; works.
In Transit	API does not emit this status today; tab typically empty. Transit still impacts MRP.
Reserved	API does not emit this status today; tab typically empty.
Non-Moving	No connected movement history/status; tab typically empty.

TABLE COLUMNS

Material Code	material_code + plant_code; that pair identifies one stock row.
Material Name / Category / Unit	From material master description, category, and uom.
Current Stock	on_hand_qty; physical/opening supply. Missing row displays zero.
Reserved	reserved_qty. Displayed; not deducted by current forward MRP.
Available Stock	max(current stock - reserved, 0), calculated by UI.
Min. Stock	constraint min_stock_qty, else inventory safety_stock_min, else material safety_stock_min, else zero.
Max. Stock	constraint max_storage_capacity, else inventory max_capacity, else material max_capacity, else zero.
Last Receipt Date	Actually mapped to in_transit_due_date (expected arrival), not historical receipt date.
Inventory Value	on_hand_qty x material unit_cost; display uses rupee sign.
Status	On hand 0 = Out; else available below minimum = Low; otherwise In Stock.
Selection / Actions / Footer	Checkboxes, action notification and page numbers are visual; footer count is client-filtered.

CSV upload: exact field reference

All headers are required. The transit due-date column must exist, but its row value may be blank.

REQUIRED HEADERS

plant_code,material_code,on_hand_qty,reserved_qty,in_transit_qty,in_transit_due_date,scheduled_receipts_qty

FIELD MEANINGS + VALIDATION

plant_code	Plant master key; must exist.
material_code	Material master key; must exist.
on_hand_qty	Required numeric >= 0; current physical stock.
reserved_qty	Required numeric >= 0 and cannot exceed on_hand_qty. UI subtracts it for Available Stock; MRP currently does no
in_transit_qty	Required numeric >= 0; inbound quantity waiting for arrival.
in_transit_due_date	Header required, value optional. If present use YYYY-MM-DD. Blank date makes MRP count transit in first bucket.
scheduled_receipts_qty	Required numeric >= 0; undated balance counted in first bucket. CSV does not load separately dated future receipt

EXAMPLE ROWS

P_VAL,PK-BAG-45,12000,500,2000,2026-12-15,300

P_VAL,RM-CLINKER,800,0,0,,0

Use UTF-8 comma-delimited CSV, exact unique headers, one row per plant/material, matching column counts, and no N/A placeholders. Excel regional date display does not change the required ISO date value.

What persists and what is a mockup?

Use database-backed CSV for real stock maintenance.

IMPORT BEHAVIOR

The complete CSV is validated in a database transaction. A failure rolls back the file. A successful import inserts or updates the tenant + plant + material record. A repeat upload replaces that that pair quantities; it does not append history. It cannot create missing plants or materials. Database quantities support four decimal places.

ADD INVENTORY FORM

The form displays material code/name/type/category/unit/description, opening/min/max/safety stock, supplier/preferred supplier, price/currency, storage/shelf, lead time, and minimum order quantity. These are design fields only: submit adds a temporary browser row; Save as Draft only notifies. Use CSV for persisted changes.

DETAIL MODAL AND TABS

Overview / Stock Details	Some values are API-backed. Safety Stock 100 Kg, Opening Stock 200 Kg, Last Issue Date are hard-coded examp
Stock Transactions / Recent Transactions	No transaction endpoint; movement list is empty.
Suppliers, Purchase History, Usage History, Related Materials	Shown, Related Materials detail data is loaded.
Stock Movement chart	Static illustrative six-month bars, not actual receipts/issues.
Status / Reorder Required / Lead Time	Status API-derived; reorder compares on-hand with min; lead time fixed at 10 days.
Edit / View All / row actions	Notification-only, no database write or history retrieval.

The "Last Receipt Date" column is currently an inbound ETA field. Read it as transit due date in this release.

How inventory changes the MRP plan

Use the bucket timing and quantity rules to validate a simple case by hand.

CONCEPTUAL BUCKET FLOW

1. Start from on_hand_qty
2. Add scheduled receipt in first bucket
3. Add transit in due-date bucket (blank = first bucket)
4. Consume gross material need
5. Compare projected stock to effective minimum
6. Recommend net order; apply MOQ, multiple, lead time
7. Flag capacity excess

IMPORTANT CALCULATION BOUNDARIES

Reserved quantity	Imported and shown but current forward MRP starts with on-hand and does not subtract reserved.
Scheduled receipts	One undated CSV total; assumed available in first bucket.
Transit due date	Selects receipt bucket; empty means first bucket.
Reorder level	Different from min stock; not part of current forward netting.
Capacity	Can create an alert; it does not cap the order quantity.
Inventory value	Estimate from quantity x unit cost, not financial ledger valuation.

OPERATOR CHECKLIST

1. Verify tenant. 2. Upload plant/material masters and optional constraints. 3. Check units and plant/material codes. 4. Upload exact CSV headers. 5. Confirm success row count. 6. Compare list values with CSV and precedence rules. 7. Run a matching MRP scenario and inspect opening stock and receipt buckets. 8. Reconcile gross need and the safety-stock gap manually.